

Review retrospective: *anharmonic third-cumulant asymptotic*

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Abstract

The second Tide-review, and the first selected automatically by `tide-review-pick`. The target was `Laplace/OneD/AnharmonicKappa3.lean` — the cross-repo bridge that derives the leading-order third cumulant of the 1D anharmonic Gibbs measure, $t^2 \kappa_3(x, x, x) \rightarrow -\alpha/\lambda^3$, from the abstract `Threepoint.kappa3` framework and the concrete Laplace asymptotics. GPT-5.5 Pro reviewed it against the Susceptibility Primer’s `\leanref`-tagged input asymptotics and the file’s retrospective, and returned a *clean fidelity pass*: it independently re-derived the historically error-prone limit constants ($-\alpha/\lambda^3$ and $-5\alpha/2\lambda^3$, the 3/2-vs-5/2 coefficient) and confirmed the cumulant identity, the trilinear *abc* factor, and the signs. One signature-preserving simplification was implemented under the review discipline (extracting a duplicated $\langle x \rangle_t \rightarrow 0$ derivation into a shared private helper), affirmed by a second GPT pass, and merged. One genuine *terminology* finding — the `*.affine.*` theorems handle scalar multiples $b \cdot x$, not affine maps $c \cdot x + d$ — was escalated to the human as a future-tide rename rather than touched here.

1 Setting

This review continues the inward-facing arc opened by the pure-quartic review.¹ It is the first review whose *target* was chosen by the new `tide-review-pick` skill, which surveys the seabeds for never-reviewed (or drifted-since-review) files and ranks them by the richness of their informal anchor — the principle being that an independent fidelity check is most valuable where there is documented ground truth to check against. The top recommendation was this file, on the strength of its anchor: the Susceptibility Primer states its three input asymptotics with `\leanref` tags, so the reviewer had the paper’s own constants to compare the Lean against.

The reviewed file sits at `laplace` commit `b7188f4`. It is a cross-repo bridge: it imports `Threepoint.kappa3` (the abstract third-cumulant framework) and the concrete anharmonic asymptotics from `IntegralRemainder.lean`, and assembles

$$t^2 \kappa_3(x, x, x) \rightarrow -\frac{\alpha}{\lambda^3}$$

for $L(x) = \frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ with $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$.

2 What was reviewed

The file’s eight declarations (three private helpers, five public theorems) build to the headline κ_3 asymptotic and its corollaries (the third-moment asymptotic and the affine / scalar-multiple

¹2026-05-31-review-2d-pure-quartic.

lift).² The informal corpus: the file’s module docstring, the file’s own retrospective,³ and — the load-bearing anchor — three primer equations, each `\leanref`-tagged to this seabed: $\langle x \rangle_t = -\alpha/(2\lambda^2 t) + O(t^{-2})$, $\text{Cov}_t[x, x] = 1/(\lambda t) + O(t^{-2})$, and $\text{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + O(t^{-3})$. The file’s potential and hypothesis triple match the primer verbatim.

3 Fidelity findings

None — a clean pass, with the arithmetic independently re-derived. The review’s highest value here was checking the *limit constants*, which combine three input asymptotics through a cumulant identity and are exactly the kind of coefficient bookkeeping that is easy to get wrong. GPT-5.5 Pro verified each step from first principles:

- the cumulant expansion $\kappa_3(x, x, x) = \langle x^3 \rangle - 3\langle x^2 \rangle \langle x \rangle + 2\langle x \rangle^3$ (coefficients 1, -3, +2);
- the headline $-\frac{2\alpha}{\lambda^3} - 2 \cdot \frac{1}{\lambda} \cdot \left(-\frac{\alpha}{2\lambda^2}\right) + 0 = -\frac{\alpha}{\lambda^3}$;
- the third moment $-\frac{2\alpha}{\lambda^3} + \frac{1}{\lambda} \left(-\frac{\alpha}{2\lambda^2}\right) = -\frac{5\alpha}{2\lambda^3}$ — explicitly flagging and confirming the historically error-prone 3/2-vs-5/2 constant;
- the trilinear *abc* factor and the sign on $-(abc)\alpha/\lambda^3$.

GPT-5.5 Pro: I do not see a mismatch in any of the public theorem statements or limit constants relative to the attached ground truth. The coefficient arithmetic that was historically error-prone (3/2 vs 5/2) is consistent in the final public results here.

A terminology finding (escalated, not fixed). The `*_affine*` theorems — and the docstrings calling them “affine multilinearity” — actually handle *scalar multiples* $b \cdot x$, not affine maps $c \cdot x + d$. No statement is wrong against its own target (the target was exactly $(b \cdot x, a \cdot x, c \cdot x)$), so this is not a fidelity defect; but the name and prose overclaim. Because renaming changes the public API, it is recorded as a future forward-tide item, not touched in this review. This is the review discipline working as intended: the reviewer surfaces a real imprecision, and the fix that would change a signature is escalated rather than slipped into a “simplification”.

4 Simplification applied

Of GPT’s five proof-body suggestions, one was implemented; four were deferred to keep the run bounded.

Landed. Both the second-moment and the headline κ_3 asymptotic proofs contained a near-identical ~ 20 -line derivation of $\langle x \rangle_t \rightarrow 0$ (as $(1/t) \cdot (t\langle x \rangle_t)$, with $1/t \rightarrow 0$). This was hoisted into a single private `mean_anharmonic_tendsto_zero` lemma that both proofs now call in one line, and the now-unused local `h_inv_t` witnesses were removed. The new lemma is *private*: the file’s five *public* theorem signatures are byte-identical before and after, which is the review discipline’s hard invariant. Full library build green, no `sorry/axiom`; the file drops from 488 to 479 lines with the duplication gone.

²Public theorems: `secondMoment_anharmonic_asymptotic`, `kappa3_anharmonic_id_id_id_asymptotic`, `thirdMoment_anharmonic_asymptotic`, `kappa3_affine_id_id_id_eq`, `kappa3_anharmonic_affine_asymptotic`.

³2026-05-06-tide-anharmonic-kappa3.

Deferred. Replacing the square lambda-collapse `haves` with `simp [pow_two]` (the mixed-product ones genuinely need `ring`); a few `.const_mul` tidies; a `let L :=` local abbreviation. All signature-preserving, all lower-yield; left for a later pass.

A second GPT-5.5 Pro pass on the diff affirmed the change (“a faithful realization ... preserves the public surface ... overall verdict: accept”) and agreed both the deferrals and the escalate-not-fix call on the naming.

5 Disagreements and open questions

None requiring adjudication: the fidelity pass was clean and the affirmation pass accepted the implemented change. The only item handed to the human is the `*_affine_*` naming, below.

6 Lessons

- *The arithmetic check is the point.* For a file whose headline is a constant assembled from several asymptotics through an algebraic identity, the single most valuable thing an independent reviewer does is re-derive that constant from scratch. The clean pass here is real evidence, from a second model, that the $-5\alpha/2\lambda^3$ did not pick up the classic off-by-a-half.
- *Naming is a fidelity surface too.* “Affine” that is really “scalar-multiple” is exactly the sort of informal/formal gap a review should catch — and exactly the sort whose fix (a rename) belongs in a forward tide, not a review. The two-bucket discipline routed it correctly without prompting.
- *Private-helper extraction is the safe simplification.* Hoisting shared proof code into a *private* lemma keeps the public surface byte-identical by construction, so it sails through the statement-invariance gate. It is becoming the workhorse review simplification (cf. the `num_zero` helper in the pure-quartic review).

7 Follow-ups

- *Forward tide (signature-changing):* rename the `*_affine_*` theorems to `*_smul_*` / `*_linear_*`, or genuinely generalise them to affine observables $c \cdot x + d$; promote the generic “ $t \cdot f(t) \rightarrow c \Rightarrow f(t) \rightarrow 0$ ” shape (now embodied by `mean_anharmonic_tendsto_zero`) and `threepoint_gibbsExp_volume_zero_eq` to `lean-common` once a second consumer appears.
- *Later review pass:* the four deferred proof-style simplifications (B2–B4).
- *Coverage:* `tide-review-pick`’s survey still lists `TwoD/SemiDegenerate` (same family as the pure-quartic review) and the J-function / harmonic-observable files as never-reviewed; a first review in a non-laplace seabed remains the coverage frontier.

Acknowledgements

GPT-5.5 Pro performed both the independent review and the affirmation pass; the target was selected by `tide-review-pick`. Full responses are in the review log.⁴

⁴`projects/primer/review-log/2026-05-31-review-anharmonic-kappa3.md`.